

NI TB Genomic Surveillance

Outbreak Investigation Report

Generated: 2026-05-08 16:05:08 UTC

Case Summary

Total Cases	250
Clustered Cases	131
Unclustered Cases	119
Open Clusters	6

Automated Interpretation Flags

- Open clusters requiring investigation: 6.
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	2000
MCMC Iterations	2500
Burn-in	500
Mean Log-Likelihood	-5250.50
Log-Likelihood SD	145.30
Transmission Probability	0.085
Generation Time Mean (days)	12.50
Generation Time SD (days)	3.200
Sampling Probability	0.650
Convergence Diagnostic	Rhat < 1.1 (good convergence)

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	32
Sequenced Cases	27
Sequencing Coverage (%)	84.38
QC Reported Cases	0
QC Pass Cases	0
QC Fail Cases	0
QC Pass Rate (%)	None
Contamination Flags	0
Median Days Specimen to QC	None

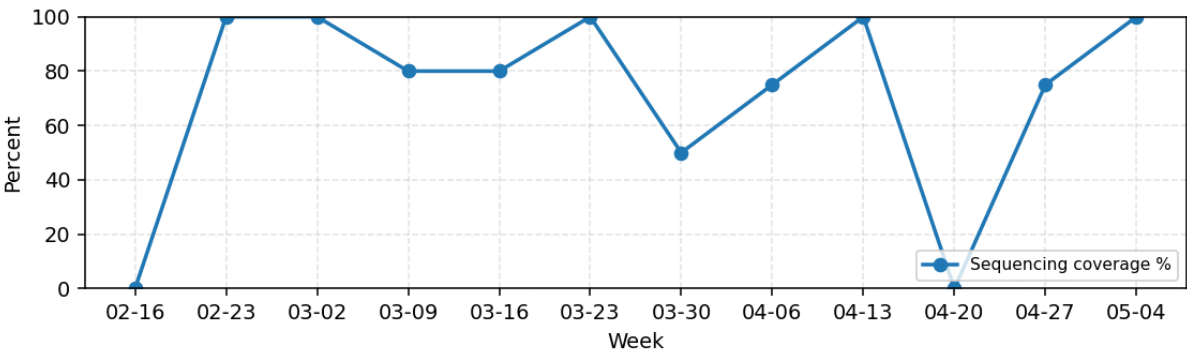
Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
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United Kingdom	32	27	84.38
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Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-16	0	0	None	None
2026-02-23	2	2	100.00	None
2026-03-02	2	2	100.00	None
2026-03-09	5	4	80.00	None
2026-03-16	5	4	80.00	None
2026-03-23	2	2	100.00	None
2026-03-30	2	1	50.00	None
2026-04-06	4	3	75.00	None
2026-04-13	2	2	100.00	None
2026-04-20	0	0	None	None
2026-04-27	4	3	75.00	None
2026-05-04	3	3	100.00	None



Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
97c6a3b2	27	1	2026-04-11	open	62
8a169590	25	1	2026-03-26	open	57
68615065	24	1	2026-04-30	open	57
edcc7ea8	21	1	2026-05-06	open	51
1de5c536	20	1	2026-05-03	open	49
b6d6762f	14	1	2026-05-07	open	37

Priority score combines cluster size, cross-region spread, specimen recency, and open investigation status.

Transmission Priority Signals

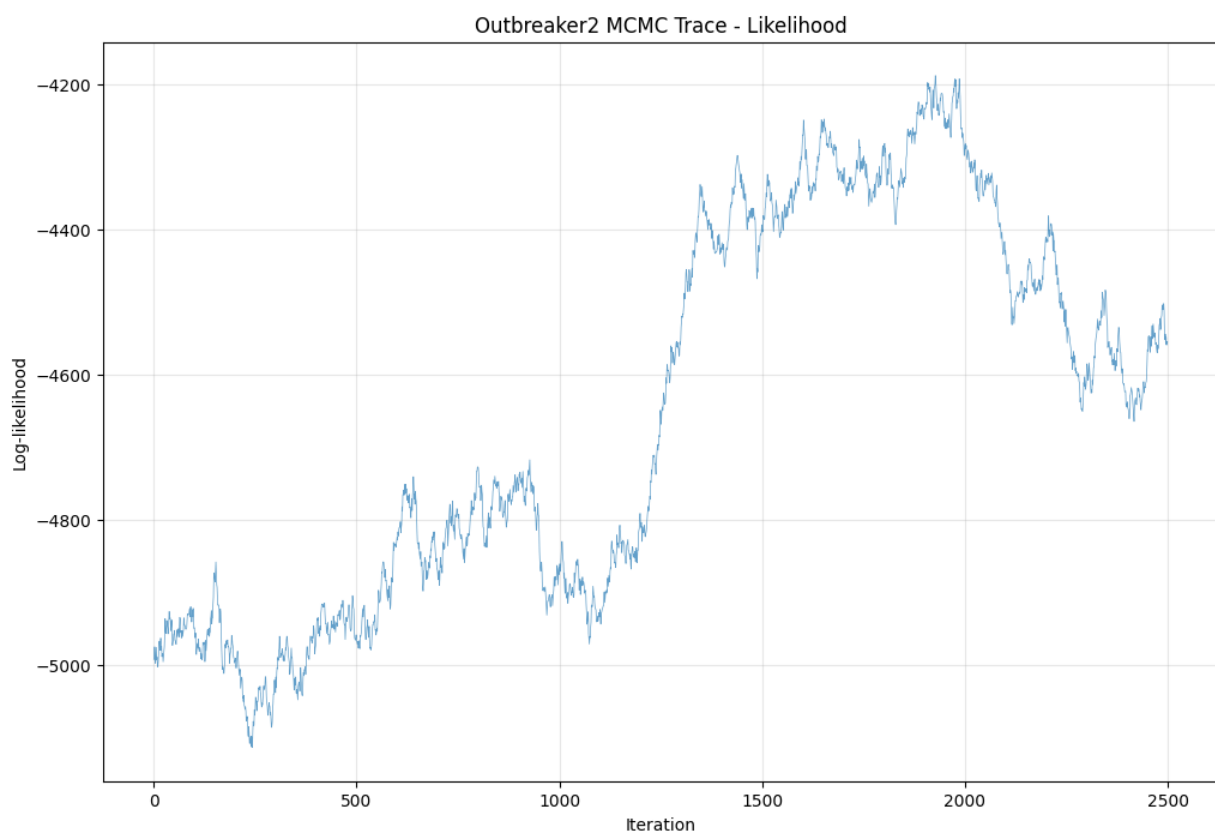
Case	Region	Risk	Out	In
ccf5b00f	United Kingdom	1.648	2	2
a53a9416	United Kingdom	1.648	2	2

1ec96485	United Kingdom	1.627	2	2
d9985b43	United Kingdom	1.621	2	2
da3c3345	United Kingdom	1.613	2	2
e439f41c	United Kingdom	1.611	2	2
3061f97f	United Kingdom	1.605	2	2
a4335f3b	United Kingdom	1.599	2	2
b5883cc4	United Kingdom	1.592	2	2
ce770af8	United Kingdom	1.591	2	2

Network snapshot: nodes=131, edges=244, high-confidence links=90

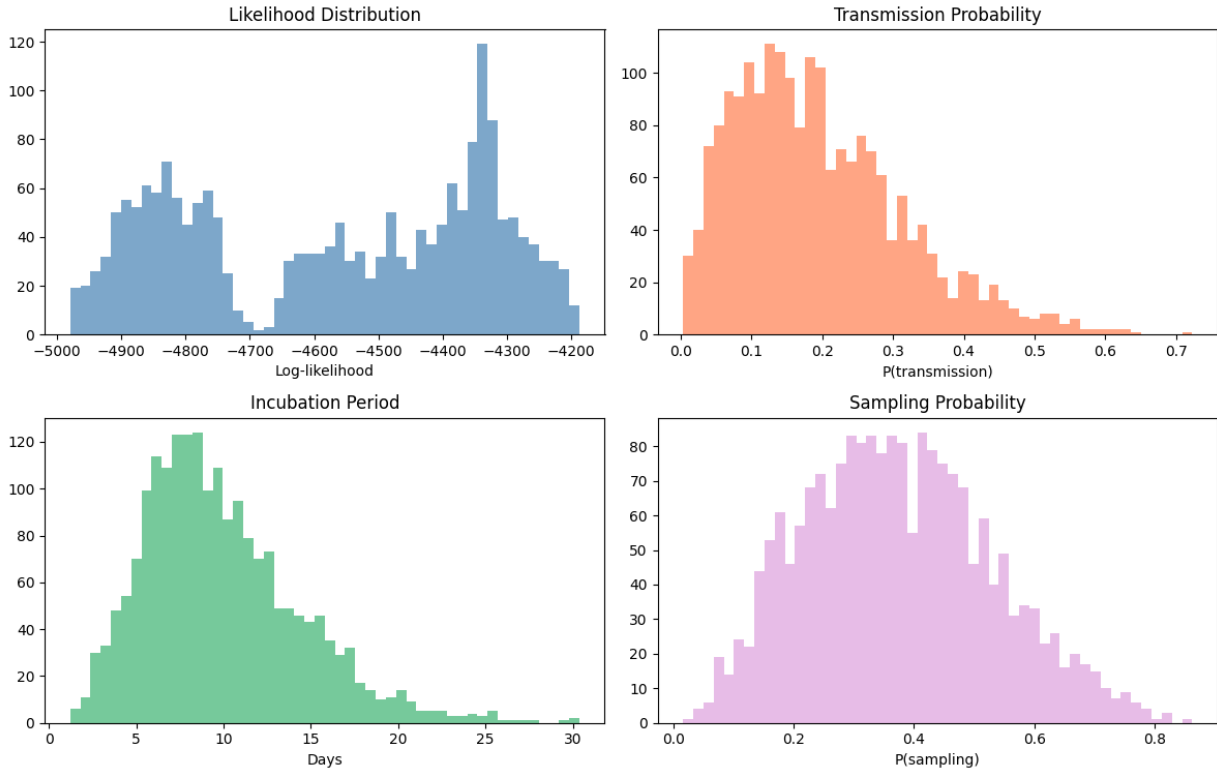
Diagnostic Graphics

Trace



Hist

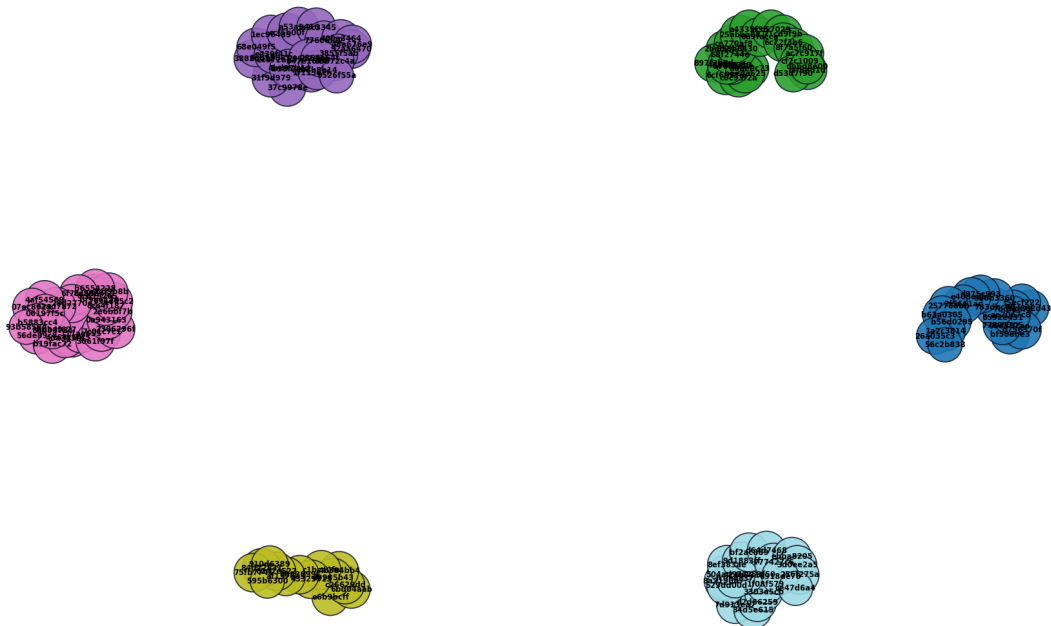
Outbreaker2 Posterior Distributions

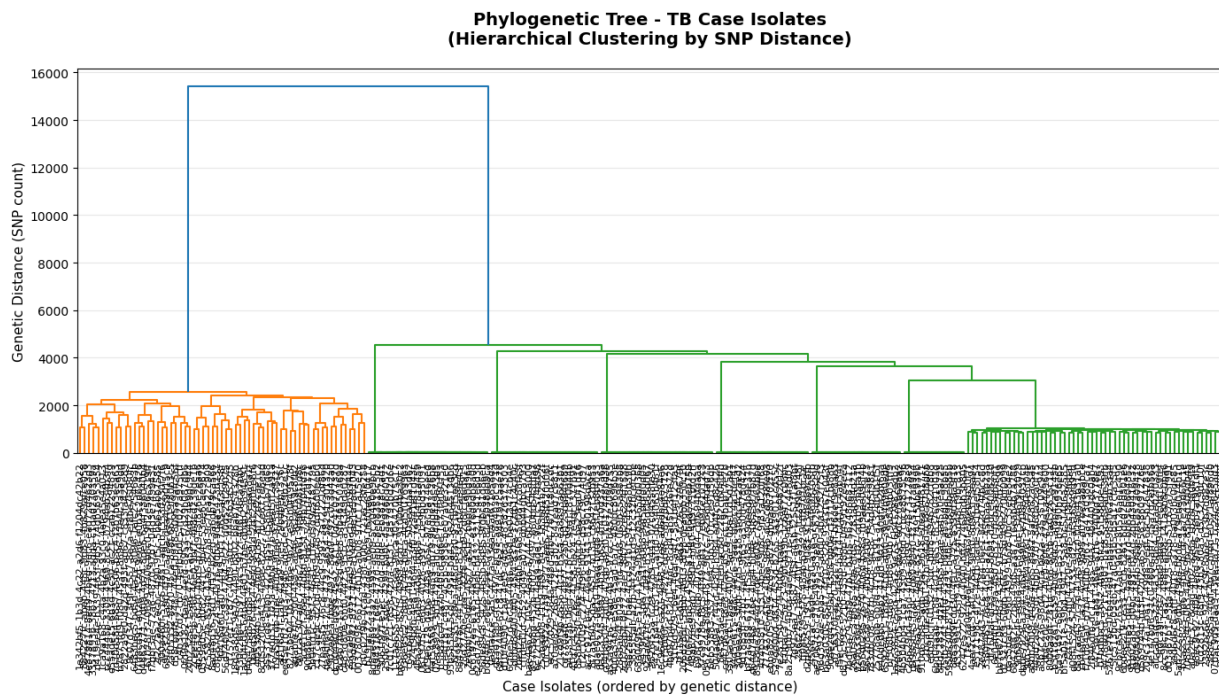


Tree

Enhanced Transmission Network

Node size = inferred spread risk | Edge label = transmission probability | Color = cluster

**Phylo**



Resistance



Data Provenance

This report combines case tables, cluster assignments, outbreaker summary metrics, surveillance KPIs, and transmission network outputs available at generation time.

Outbreaker summary timestamp: 2026-05-08T17:04:24.401249

Transmission network timestamp: 2026-05-08T16:04:40